# Common Probabiilty Distributions

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# Outline

1. Probability Distribution
2. Some Common Discrete Probability Distributions
3. Some Common Continuous Probability Distributions

# Probability Distribution

# Probability Distribution

- A variable which takes numerical values resulting from random experiments, is called a random variable (RV).
- Recall that probability distribution means the distribution of the probabilities among the different values of a random variable

There are some distributions that often arise frequently in real life. This slide shall discuss a few such distributions

# Types of Probability Distribution

Recall that depending on the type of variable, distributions are of two types:

- **Discrete probability distribution**: probability distribution of a discrete random variable
- **Continuous probability distribution**: probability distribution of a continuous random variable

# Probability Mass/Density Functions

- The probability distribution of a discrete random variable is given by its probability mass function (PMF)
  - Denoted by $p(x)$
  - $p(x) = P(X = x)$
- The probability distribution of a continuous random variable is given by its probability density function (PDF)
  - Denoted by $f(x)$
  - $\int_a^b f(x)\,dx = P(a < X < b)$

Only for the continuous case: $P(X = k) = 0$, where $k$ is some constant.

# Cumulative Distribution Function (CDF)

- For discrete random variable, $F(x) = P(X \leqslant x) = \sum_{k=-\infty}^{x} p(k)$
- For continuous case, $F(x) = P(X \leqslant x) = P(X < x) = \int_{-\infty}^{x} f(x)\,dx$
- It is a non-decreasing function, the largest value $F(x)$ can take is $1$

Here, summing/integrating from $-\infty$ means that the lower limit of the sum/integral is the smallest value in the domain of the random variable $X$.

# Some Common Discrete Probability Distributions

# Some Common Discrete Probability Distributions

## Bernoulli and Binomial Distributions

# Bernoulli Process

When a random process or experiment, called a trial, can result in only one of two mutually exclusive outcomes, such as dead or alive, sick or well, full-term or premature, the trial is called a Bernoulli trial.

- Each trial results in one of two possible, mutually exclusive, outcomes. One of the possible outcomes is denoted (arbitrarily) as a success, and the other is denoted a failure
- The probability of a success, denoted by $p$, remains constant from trial to trial. The probability of a failure, $1 - p$, is denoted by $q$
- The trials are independent; that is, the outcome of any particular trial is not affected by the outcome of any other trial

# Bernoulli Distribution

Let $X$ be the number of successe in a single Bernoulli trial. Then $X$ follows a Bernoulli distribution. It is written as:

$$X \sim \mathrm{Bernoulli}(p).$$

The probability mass function (PMF) of $X$ is given as:

$$p(x) = p^x (1-p)^{1-x}\,; \quad x = 0, 1$$

- Expected value, $\mathrm{E}(X) = 1 \times p + 0 \times (1-p) = p$
- Variance:

$$\begin{aligned} \mathrm{V}(X) = \mathrm{E}(X^2) - [\mathrm{E}(X)]^2 &= (1^2 \times p + 0^2 \times (1-p)) - p^2 \\ &= p - p^2 = p(1-p) \end{aligned}$$

# Binomial Distribution

Let $X$ be the number of successes in $n$ Bernoulli trials, where the probability of success is $p$. Then the random variable $X$ follows a binomial distribution. It is written as:

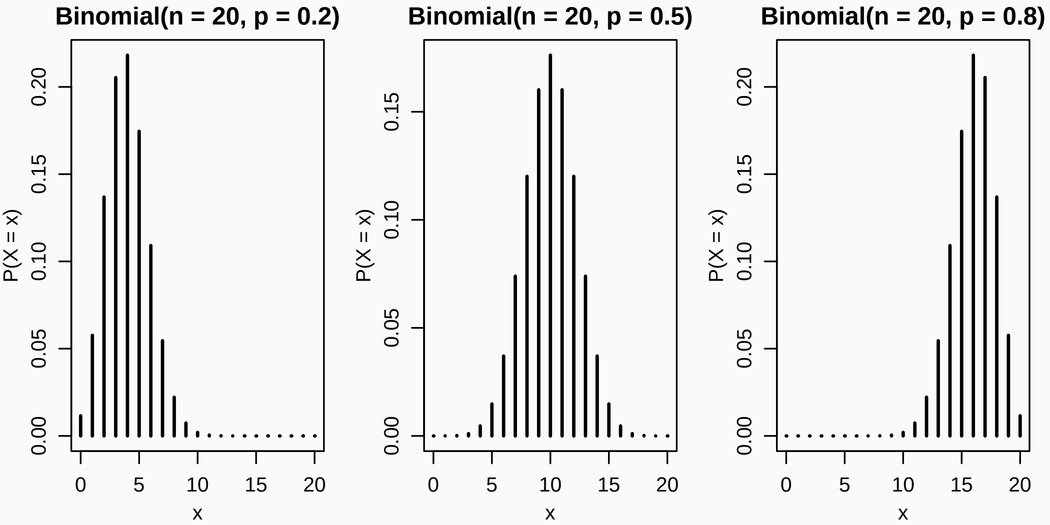
$$X \sim \mathrm{binom}(n, p).$$

The probability mass function (PMF) of $X$ is given as:

$$p(x) =^{n} \mathrm{C}_{x}\; p^{x}(1-p)^{n-x}\,; \quad x = 0, 1, 2, 3...$$

- $\mathrm{E}(X) = np$
- $\mathrm{V}(X) = np(1-p)$

# Binomial Distribution: Shape



# Examples of Binomial Variables

- Number of defective products among a batch of products
- Number of patients carrying a ceratain pathogen among a group of patients
- Number of heads when a coin is tossed several times

# Binomial Distribution: Example

There are 3 multiple choice questions in a MCQ test. Each MCQ consists of four possible choices and only one of them is correct. If an examinee answers those MCQ randomly (without knowing the correct answers).

1. What is the probability that exactly any two of the answers will be correct?
2. What is the probability that at least two of the answers will be correct?
3. What is the probability that at most two of the answers will be correct?
4. What will be the average or expected number of correct answers?
5. Also, find the standard deviation of number of correct answers.

# Binomial Distribution: Example Solution

Each MCQ has $4$ options. So, when answering randomly, probability of correct answer, $p = 1/4 = 0.25$. Let the random variable $X$ denote the number of correct answers. Then, $X \sim \text{binomial}(3, 0.25)$.

1. Probability of two correct answers,
   $P(X = 2) = {}^3\mathrm{C}_2\; 0.25^2\,(1 - 0.25)^{3-2}$
2. Probability of *at least* two correct answers $= P(X = 2) + P(X = 3)$
3. Probability of *at most* two correct answers
   $= P(X = 0) + P(X = 1) + P(X = 2)$
4. Expected number of correct answers, $\mathrm{E}(X) = 3 \times 0.25 = 0.75$
5. Variance of number of correct answers, $\mathrm{V}(X) = 3 \times 0.25 \times (1 - 0.25)$

The calculations for question 2 and 3, are left as exercises.

# Some Common Discrete Probability Distributions

## Poisson Distribution

# Poisson Distribution

A Poisson random variable is a discrete random variable that counts the number of times a certain event occurs within fixed interval of time, or area, or volume or any other unit.

The Poisson distribution describes the probability of these counts.

Let $X \sim \text{Poisson}(\lambda)$, then the PMF of $X$ is given by:

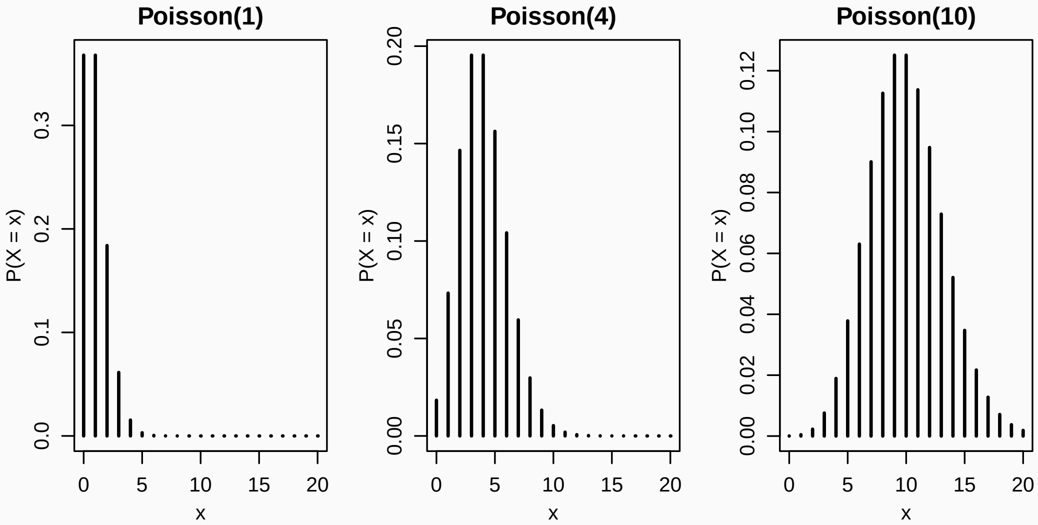
$$P(X = x) = \frac{e^{-\lambda}\lambda^{x}}{x!}; \quad \lambda > 0; \quad x = 0, 1, 2, 3, \ldots$$

- $\lambda$ is the parameter of the distribution
- $\mathrm{E}(X) = \lambda$
- $\mathrm{V}(X) = \lambda$

# Examples of Poisson Random Variables

- Number of phone calls received by a call center in one minute
- Number of typing errors on a page of text
- Number of accidents at an intersection in a day
- Number of emails arriving in an hour

# Poisson Distribution: Shape



# Assumptions of a Poisson Process

- Events occur independently of each other
- The probability of more than one event in a very small interval is negligible
- The average rate of occurrence $\lambda$ is constant over the interval
- The probability of an event occurring is proportional to the length of the interval

# Poisson Distribution: Example

The average number of errors on a page of a certain magazine is $0.2$. What is the probability that:

1. a randomly selected page contains $0$ (zero) error?
2. a randomly selected page contains $2$ or more errors?
3. two consecutive pages have $3$ errors?
4. What is the average error per page?
5. Also, find standard deviation of the number of errors in a page.

# Poisson Distribution: Example Solution

Here, error rate, $\lambda = 0.2$.

1. Probability that a randomly selected page contains $0$ errors is

$$P(X = 0) = \frac{e^{-0.2}\,0.2^0}{0!}$$

2. Probability that a randomly selected page contains $2$ or more errors is

$$\begin{aligned} P(X \geqslant 2) &= 1 - P(X < 2) \\ &= 1 - [P(X = 0) + P(X = 1)] \end{aligned}$$

The above calculations are left as an exercise.

# Poisson Distribution: Example Solution (cont.)

3. Probability that two consecutive pages have $3$ errors:
   The error rate for two pages is, $\lambda_{\text{new}} = 0.2 \times 2 = 0.4$
   Therefore, the desired probability is:

$$\frac{e^{-0.4}\, 0.4^3}{3!}.$$

4. Average error per page is $\mathrm{E}(X) = \lambda = 0.2$.
5. Standard deviation of number of errors in a page,
   $\mathrm{SD}(X) = \sqrt{\mathrm{V}(X)} = \sqrt{0.2}$.

# Some Common Continuous Probability Distributions

# Some Common Continuous Probability Distributions

## Uniform Distribution

# Uniform Distribution

A random variable with sample space $[a, b]$ and equal probability for all intervals of equal length, is called a uniform random variable.

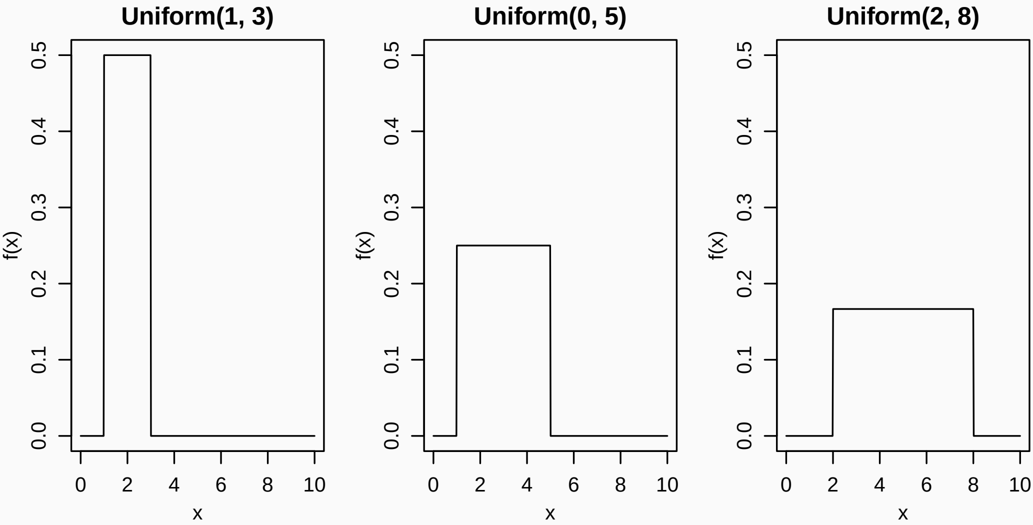
Let $X$ be a uniformly distributed random variable in $[a, b]$, it is written mathematically as $X \sim \mathrm{uniform}(a, b)$.

The PDF of $X$ is given by:

$$f(x) = \frac{1}{b-a}; \quad a \leqslant x \leqslant b$$

- $\mathrm{E}(X) = \frac{b+a}{2}$
- $\mathrm{V}(X) = \frac{(b-a)^2}{12}$

# Uniform Distribution: Shape



# Uniform Distribution: Example 1

The waiting time (in minutes) for train is uniform$(10, 50)$. Find the following:

1. The probability that you have to wait at least $20$ minutes
2. Average waiting time
3. Standard deviation of waiting time

# Uniform Distribution: Example 1 Solution

Here, wait time, $X \sim \mathrm{uniform}(10, 50)$

1. Probability of waiting at least 20 minutes:

$$P(X > 20) = P(20 \leqslant X \leqslant 50) = \int_{20}^{50} \frac{1}{50-10}\, dx$$
$$= \left[\frac{x}{40}\right]_{20}^{50} = 30/40$$

2. Average waiting time, $\mathrm{E}(X) = \frac{10+50}{2} = 30$

Question 3 is left as an exercise.

# Uniform Distribution: Example 2

Find the CDF of $X \sim \mathrm{uniform}(a, b)$. Then find the CDF the train station waiting example in page 21, i.e., when $a = 10$ and $b = 50$. Using this CDF, find the following probabilites:

1. Probability that you have to wait less than $25$ minutes
2. Probablity that you have to wait more than 25 minutes
3. Probability that the wait time is between 26 and 36 minutes
4. Probability that the wait time is less than 32 minutes or more than 42 minutes

# Uniform Distribution: Example 2 Solution

The CDF is given by:

$$
\begin{aligned}
F(X) = P(X < x) &= \int_a^x f(x)\,dx \\
&= \int_a^x \frac{1}{b-a}\,dx \\
&= \left[\frac{x}{b-a}\right]_a^x \\
&= \frac{x-a}{b-a}
\end{aligned}
$$

# Uniform Distribution: Example 2 Solution (cont.)

From the previous page: $F(x) = \frac{x-a}{b-a}$.

When $a = 10, b = 50$: $F(x) = \frac{x-10}{50-10} = \frac{x-10}{40}$.

1. $P(X < 25) = F(25) = \frac{25-10}{40} = 15/40$
2. $P(X > 25) = 1 - F(25)$
3. $P(26 < X < 36) = F(36) - F(26)$
4. $P(X < 32 \cup X > 42) = P(X < 32) + P(X > 42) = F(32) + 1 - F(42)$

# Uniform Distribution: Example 3

Derive the expected value of the uniform distribution:

$$
\begin{aligned}
E(X) &= \int_a^b x \cdot \frac{1}{b-a} \mathrm{d}x \\
&= \frac{1}{b-a} \int_a^b x \mathrm{d}x \\
&= \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b \\
&= \frac{1}{b-a} \cdot \frac{b^2 - a^2}{2} \\
&= \frac{b+a}{2}
\end{aligned}
$$

# Some Common Continuous Probability Distributions

## Exponential Distribution

# Exponential Distribution

Let $X$ be a continuous random variable measuring the waiting time until an event occurs. The event is assumed to occur randomly but at a constant average rate. And let $\lambda$ be the average number of events in one unit of time.

Then $X$ is said to follow the exponential distribution with rate parameter $\lambda$, written as $X \sim \text{exponential}(\lambda)$.

The PDF of $X$ is given by

$$f(x) = \lambda e^{-\lambda x}\,; \quad x > 0, \quad \lambda > 0.$$

- $E(X) = 1/\lambda$, it denotes the average waiting time between events
- $\mathrm{V}(X) = 1/\lambda^2$

# Examples of Exponentially Distributed Random Variables

- Waiting time until the next customer arrives at a service counter
- Time between two consecutive phone calls arriving at a call center
- Lifetime of an electronic component assuming a constant failure rate
- Time until a radioactive atom decays
- Waiting time until the next bus arrives when arrivals occur randomly
- Time between failures of a machine operating under steady conditions

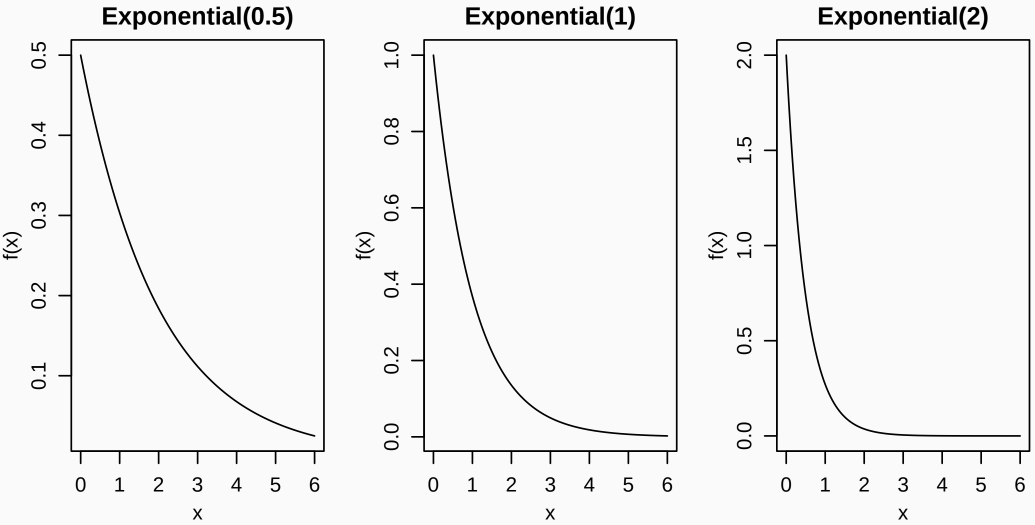
# Exponential Distribution and Poisson Distribution

If events occur randomly and independently with but at a constant average rate, $\lambda$, then:

- The number of events in a fixed time interval follows a Poisson distribution
- The waiting time between those events follows an exponential distribution

Here, $\lambda$ is the number of events that occur in one unit of time.

# Exponential Distribution Shape



# Exponential Distribution: Example 1

Average time required to repair a machine is $0.5$ hours. What is the probability that the next repair will take more than $2$ hours?

Let, $X$ = time required to repair the machine
Average time required to repare a machine:

$$\mathrm{E}(X) = \frac{1}{\lambda} = 0.5 \Rightarrow \lambda = 2$$

Therefore, $X \sim \text{exponential}(2)$
Now, Probability that the next repair will take more than two hours:

$$P(X > 2) = \int_2^\infty 2e^{-2x}dx = \left[-e^{-2x}\right]_2^\infty = \left[-0 + e^{2\times 2}\right] = e^{-4} = 0.0183.$$

# Exponential Distribuion: Exercise

Find the CDF of the exponetial distribution.

# Some Common Continuous Probability Distributions

## Normal Distribution

# Normal Distribution

A continuous random variable whose distribution is symmetric and bell shaped around its mean is called a normal random variable.

Let $X$ be a normally distributed random variable with mean $\mu$ and variance $\sigma^2$. It is written as

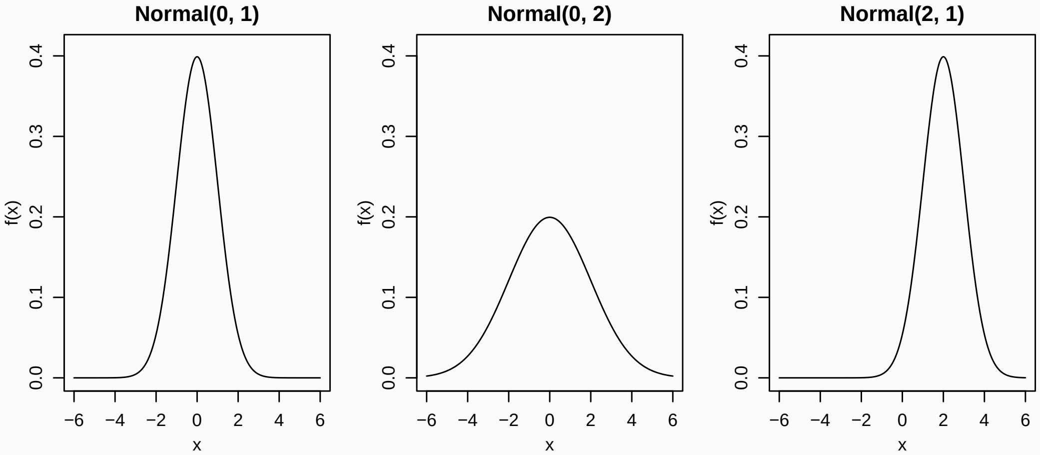
$$X \sim \mathrm{N}(\mu, \sigma^2)$$

The probability density function of $X$ is

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\Big[-\frac{(x-\mu)^2}{2\sigma^2}\Big]; \quad -\infty < x < \infty$$

- $\mathrm{E}(X) = \mu$
- $\mathrm{V}(X) = \sigma^2$

# Normal Distribution: Shape



# Properties of Normal Distribution

- The distribution is symmetric around the mean $\mu$
- Mean, median and mode are equal
- The spread of the distribution depends on the standard deviation $\sigma$
- Larger $\sigma$ produces a wider and flatter curve
- Total area under the curve is equal to $1$

# Standard Normal Distribution

A special case of the normal distribution occurs when the mean is $0$ and the standard deviation is $1$.

If $X \sim \mathrm{N}(\mu, \sigma^2)$, then the standardized variable

$$Z = \frac{X - \mu}{\sigma}$$

follows the standard normal distribution.

The standard normal distribution is written as

$$Z \sim \mathrm{N}(0, 1)$$

***Probabilities for normal variables are usually obtained using standard normal table (Z-table).***

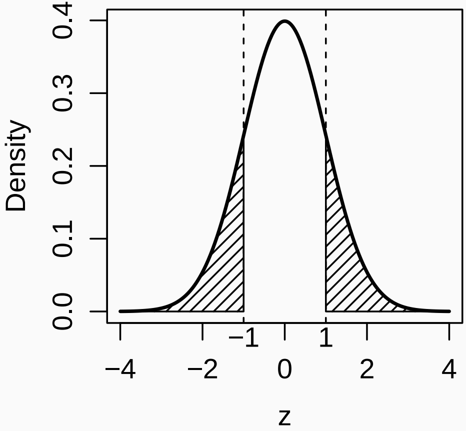
# Symmetry of the Standard Normal Distribution

The normal distribution is symmetric around the mean (which is 0 for standard normal distribution). Therefore the following relationship holds:

$$P(Z < -k) = P(Z > k)$$

where $k$ is a positive real number.

The area under the curve of the highlighted regions are equal.



# CDF of the Standard Normal Variable

- For a normal random variable $X \sim \mathrm{N}(\mu, \sigma^2)$, probabilities are usually calculated using the standard normal distribution CDF
- The cumulative distribution function (CDF) of the standard normal distribution denoted by $\Phi(x)$ is a function that gives the probability: $P(Z < z)$
- The Z-table provides the precalculated cumulative probability: $\Phi(z) = P(Z \leqslant z)$, which represents the area under the standard normal curve to the **left** of $z$

# Common Probability Intervals

The following types of probability regions commonly appear when working with the normal distribution.

Once the value of $Z$ is obtained, probabilities can be computed using simple rules.

- $P(Z < z) = \Phi(z)$
- $P(Z > z) = 1 - P(Z < z) = 1 - \Phi(z)$
- $P(a < Z < b) = P(Z < b) - P(Z < a) = \Phi(b) - \Phi(a)$
- $P(Z < a \cup Z > b) = P(Z < a) + 1 - P(Z < b) = \Phi(a) + 1 - \Phi(b)$
- $P(Z > k) = P(Z < -k)$, where $k$ is a positive real number

# Example: Finding Probability of Normal Random Variable

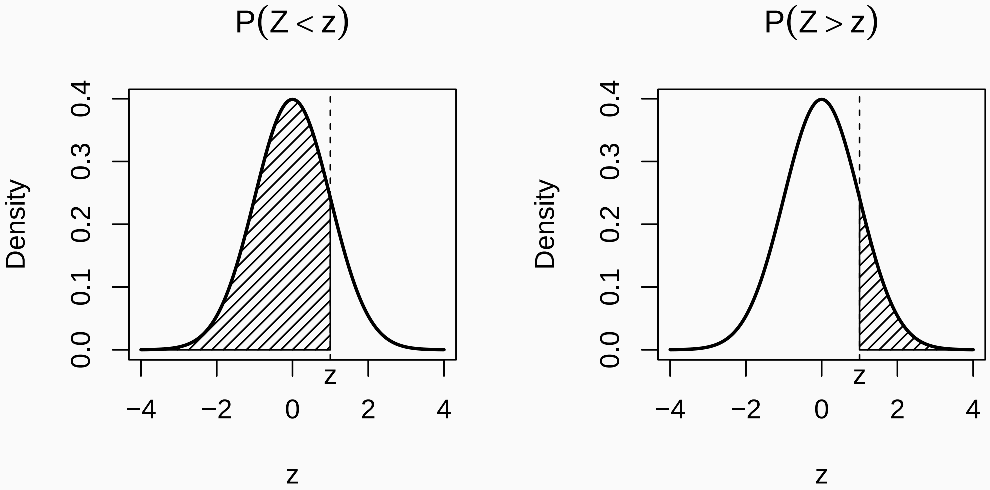
Let a random variable $X \sim \mathrm{N}(10, 25)$. Find $P(X < 4)$ and $P(X > 16)$:

$$P(X < 4) = P\left(\frac{X - 10}{\sqrt{25}} < \frac{4 - 10}{\sqrt{25}}\right) = P(Z < -1.2) = \Phi(-1.2) = 0.115$$

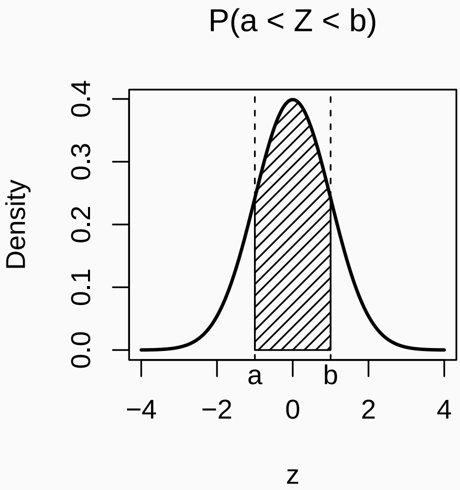
$$\begin{aligned} P(X > 16) = P\left(\frac{X - 10}{\sqrt{25}} > \frac{16 - 10}{\sqrt{25}}\right) = P(Z > 1.2) &= 1 - P(Z < 1.2) \\ &= 1 - \Phi(1.2) \\ &= 1 - 0.885 \\ &= 0.115 \end{aligned}$$

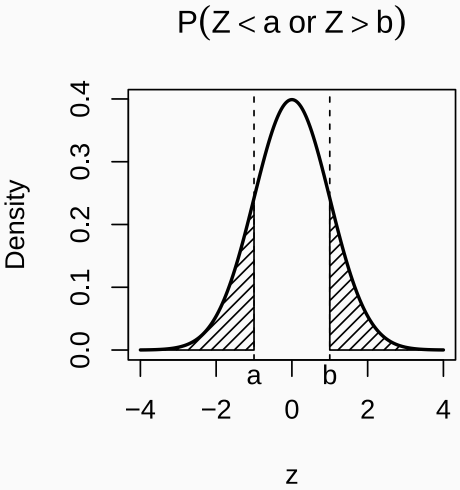
The above values of $\Phi(\cdot)$ are found using the Z-table.

# Normal Distribution: Probability Regions



# Normal Distribution: Probability Regions (cont.)



# Normal Distribution: Example

The heights of students in a university are normally distributed with mean $170$ cm and standard deviation $10$ cm.

Find the following:

1. The probability that a randomly selected student is taller than $180$ cm
2. The probability that a student's height lies between $160$ cm and $180$ cm
3. The average height of students

# Normal Distribution: Example Solution

Let height $X \sim \mathrm{N}(170, 10^2)$

1. Probability that a student is taller than $180$ cm:

$$Z = \frac{180 - 170}{10} = 1$$

$$P(X > 180) = P(Z > 1) = 0.1587$$

2. Probability that height lies between $160$ and $180$:

$$Z_1 = \frac{160 - 170}{10} = -1$$

$$Z_2 = \frac{180 - 170}{10} = 1$$

$$P(160 < X < 180) = P(-1 < Z < 1) = 0.6826$$

# Thank you.